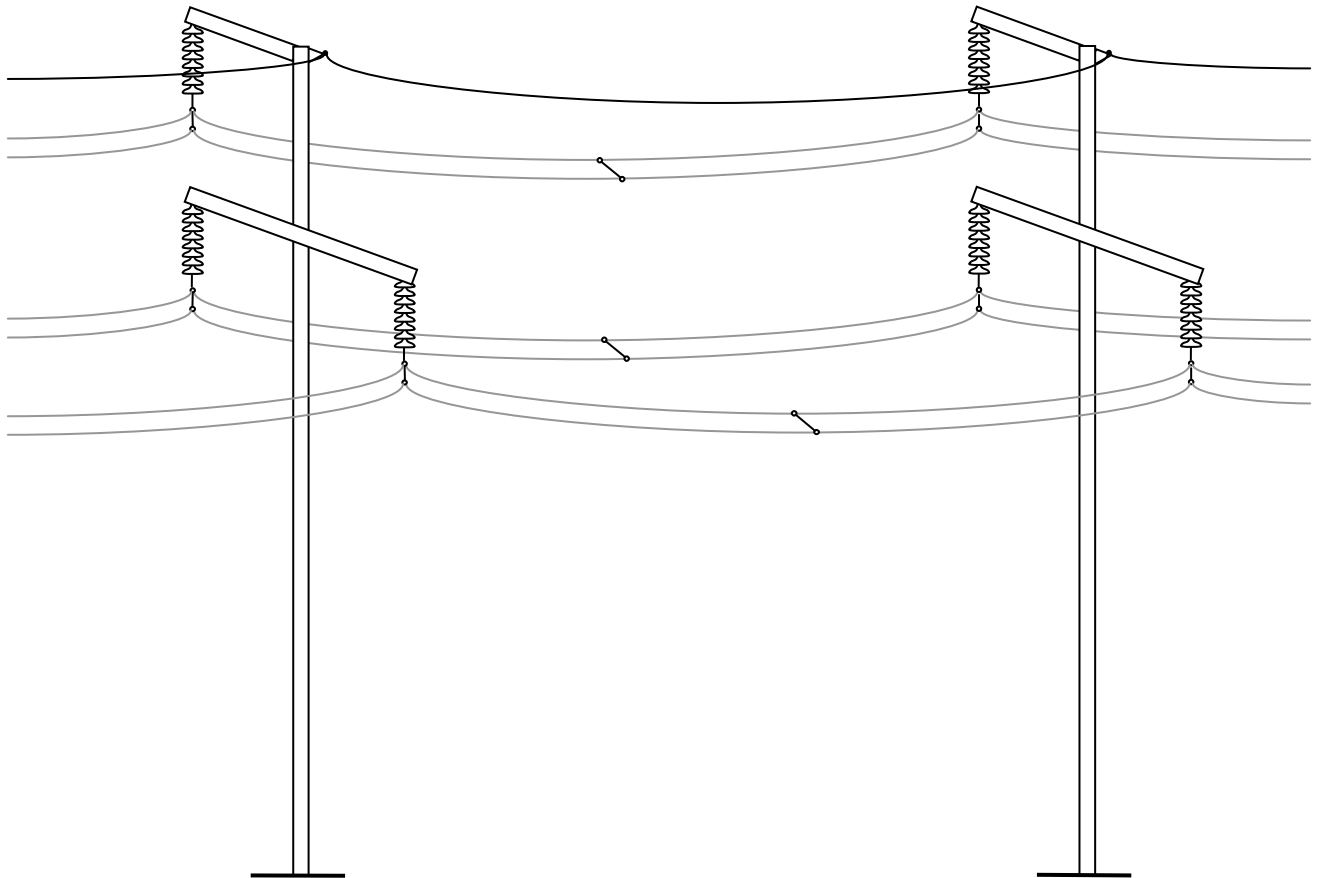


Power system transmission lines are special interconnect components used to connect parts of a power system together, usually over long distances, and while carrying large amounts of energy very efficiently. In real-world power systems, there are two basic types of transmission lines; **open conductor overhead transmission lines** and **insulated conductor power cables**.

In **open conductor overhead transmission lines**, conductors do not have an insulating coating on them. They are bare and exposed. They are positioned high above the ground to avoid human contact and spaced by relatively large distances using support towers to minimize electromagnetic effects. The energized conductors are insulated from one another through the air and through support insulators. The diagram below shows a profile view for a typical open conductor overhead 3 ϕ transmission line.



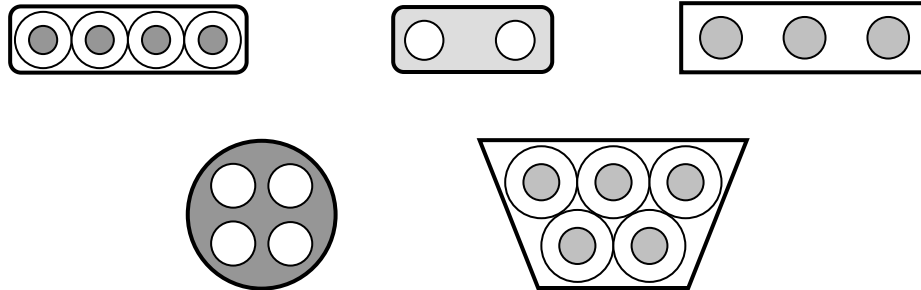
Supporting towers are used to keep the energized conductors a safe distance away from the ground. They are constructed using high strength steel and/or wooden parts. A smaller wire called a shield wire (or static wire) is attached to the top of the supporting towers. It is connected to earth ground to attract lightning strikes away from the energy carrying conductors. Conductors are electrically isolated from the structure using specially designed insulators made of ceramic or porcelain material. Insulators are built using a chain of individual discs, where the number of discs in the chain depends on the operating voltage of the line. The higher the voltage is, the longer the required insulator chain will be and also the higher above ground the conductors will be.

Transmission lines can use more than one sub-conductor per conductor. This is called **conductor bundling**. In the example above, two cables per phase are used. When conductor bundling is used, physical spacers are installed along the transmission line length to maintain a desired separation distance between the individual sub-conductors of the bundled conductor. Also the individual cables are connected to each other on each end of the transmission line in order to electrically connect the sub-conductors in parallel.

The construction costs of open conductor overhead transmission lines are in excess of \$300,000/mile for 115KV, \$600,000/mile for 230KV, and \$1,000,000/mile for 500KV. Building new open conductor overhead transmission lines are huge multi-million dollar projects that take a lot of planning, detailed engineering design, material acquisition, and construction effort.

Insulated conductor power cables are very different in their physical construction compared to open conductor overhead transmission lines. Insulated conductor power cables have an insulating coating on each conductor, and the insulated conductors are positioned closely together to form a package with minimal cross sectional area. An outer jacket or sheath is used to keep the individually insulated conductors together and to protect the insulated conductors from physical damage. Because the conductors are insulated, there is no electrical shock hazard, and direct physical human contact with the cable is possible.

Insulated conductor power cables can be run underground, underwater, and through confined physical spaces. The most common application of insulated conductor power cables is the electrical wiring in buildings where they are run inside walls. The diagrams below show physical cross sections of some possible types of insulated conductor power cables.

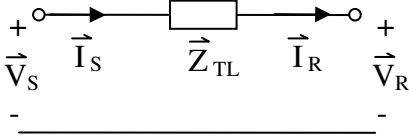
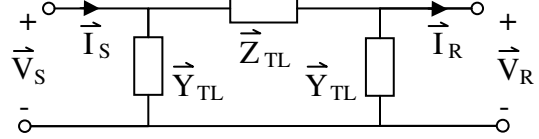


When a power system transmission line is energized by a voltage and current flows through its conductors, several electrical phenomena occur. The four most important of these phenomena are listed below.

1. **Series Resistance** inside the conductors along the length of the transmission line causes **heating of the conductors and power loss** when current flows through the transmission line conductors.
2. **Shunt Resistance** between the transmission line conductors through the insulation medium along the length of the transmission line causes **heating of the insulation medium and power loss** when voltage is applied between the transmission line conductors. This obviously is more of a concern with insulated conductor power cables which have the conductors in close proximity to one another and a solid insulation medium between the conductors. In open conductor overhead transmission lines, this phenomenon is ignored.
3. **Magnetic Field Interaction** between the transmission line conductors causes **induced voltages on the conductors** along the length of the transmission line. These induced voltages are referred to as “back emf” or “kick-back” voltages because their polarity opposes the direction of current flow in the transmission line.
4. **Electric Field Interaction** between the transmission line conductors causes **induced voltages between the conductors** along the length of the transmission line. These induced voltages have the same polarity as the voltage applied to the transmission line conductors, and they can cause the magnitude of the voltage between the conductors to increase along the length of the transmission line.

These four phenomena exist simultaneously and interact with each other to create a steady state operating condition. The physical construction of the transmission line determines how prevalent each phenomenon is to the overall electrical behavior of the transmission line. Physical construction details like conductor size, conductor material, spacing between conductors, and insulation medium all play a vital role in determining the transmission line electrical behavior.

Mathematical modeling of the detailed transmission line phenomena listed above is a complicated subject that is best handled in a course on electromagnetics. In a power system course like this one, we are more concerned with doing basic power system modeling and power system calculations. Therefore the following transmission line equivalent circuit models will just be defined for our use and will not be mathematically proven.

Transmission Line Series Impedance Model Equivalent Circuit and Parameter Definition	
	$\begin{cases} \vec{Z} = R + j120\pi L \\ \vec{Z}_{TL} = \vec{Z} \ell \end{cases} \quad \ell = \text{transmission line length}$
Transmission Line Pi Model Equivalent Circuit and Parameter Definition	
	$\begin{cases} \vec{Z} = R + j120\pi L \\ \vec{Z}_{TL} = \vec{Z} \ell \end{cases} \quad \begin{cases} \vec{Y} = G + j120\pi C \\ \vec{Y}_{TL} = \frac{1}{2} \vec{Y} \ell \end{cases}$

Notice the four variables R , L , G , C in the equations that define the circuit parameters for the transmission line models. These variables each mathematically represent one of the four previously listed transmission line phenomena. The calculation of these parameters differs between single phase and three phase transmission lines. In a single phase transmission line, there are two conductors and both of these conductors must be modeled. However in a three phase transmission line only one of the three phase conductors must be modeled since balance and symmetry are assumed and the equivalent circuits are L-N equivalent circuits. Each of these four variables and how they are calculated will next be defined in detail.

The variable R in the transmission line equivalent circuits is called the **distributed series resistance**, and it models the series resistance phenomena of the transmission line conductors. Equations for both single phase and three phase transmission lines are given below.

Series Resistance of a Single Phase Transmission Line	
$R = \left(\frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y} \right) \quad \Omega/\text{unit length} \quad R_T = R \ell \quad \Omega$	
R = distributed series resistance of the transmission line R_T = total series resistance of the transmission line n_x, n_y = number of subconductors in the two conductors R_{sx}, R_{sy} = distributed series resistance of a subconductor in each conductor ℓ = transmission line length	
Series Resistance of a Three Phase Transmission Line	
$R = \left(\frac{R_s}{n} \right) \quad \Omega/\text{unit length} \quad R_T = R \ell \quad \Omega$	
R = distributed per phase series resistance of the transmission line R_T = total per phase series resistance of the transmission line n = number of subconductors per phase conductor R_s = distributed series resistance of a subconductor in one of the phase conductors	

The variable L in the transmission line equivalent circuits is called the **distributed series inductance**, and it models the magnetic field interaction phenomena between the transmission line conductors through the insulation medium. Equations for both single phase and three phase transmission lines are given below.

Series Inductance of a Single Phase Transmission Line

$$L = k_{L1} \cdot \ln \left(\frac{GMD_{xy}}{\sqrt{GMR_x GMR_y}} \right) \quad k_{L1} = 6.437376 \times 10^{-4} H / mile \quad L_T = L \ell \quad H$$

$$= 4 \times 10^{-7} H / meter$$

L = distributed series inductance of the transmission line

L_T = total series inductance of the transmission line

$$GMD_{xy} = \left[\prod_{i=1}^{n_x} \left(\prod_{j=1}^{n_y} D_{xij} \right) \right]^{\frac{1}{n_x n_y}} \quad GMR_x = \left[\prod_{i=1}^{n_x} \left(\prod_{j=1}^{n_x} D_{xij} \right) \right]^{\frac{1}{n_x^2}} \quad GMR_y = \left[\prod_{i=1}^{n_y} \left(\prod_{j=1}^{n_y} D_{yij} \right) \right]^{\frac{1}{n_y^2}}$$

GMD_{xy} = geometric mean distance between conductor x and conductor y

GMR_x, GMR_y = geometric mean radii of the two conductors

D_{xij} = distance between subconductor i in conductor x and subconductor j in conductor y

D_{xix} = distance between subconductor i in conductor x and subconductor j in conductor x

D_{yij} = distance between subconductor i in conductor y and subconductor j in conductor y

if $i = j$ in D_{xix} or D_{yiy} , then the quantity is called the self distance D_{ii} of subconductor i and is defined:

$D_{ii} = r_s e^{-0.25}$ if the subconductor is a solid cable, where r_s is the cable radius

$D_{ii} = GMR_s$ if the subconductor is a stranded cable, where GMR_s is the cable GMR from a table

Series Inductance of a Three Phase Transmission Line

$$L = k_{L3} \cdot \ln \left(\frac{GMD_{eq}}{GMR} \right) \quad k_{L3} = 3.218688 \times 10^{-4} H / mile \quad L_T = L \ell \quad H$$

$$= 2 \times 10^{-7} H / meter$$

L = distributed per phase series inductance of the transmission line

L_T = total per phase series inductance of the transmission line

$$GMD_{eq} = \sqrt[3]{D_{AB} D_{BC} D_{CA}} \quad GMR = \left[\prod_{i=1}^n \left(\prod_{j=1}^n D_{ij} \right) \right]^{\frac{1}{n^2}}$$

GMD_{eq} = geometric mean distance between the phase conductors

D_{AB}, D_{BC}, D_{CA} = distances between the three phase conductor centers (or a common point in all three)

GMR = geometric mean radius of one of the phase conductors

D_{ij} = distance between subconductor i and subconductor j in a phase conductor

if $i = j$ in D_{ij} , then the quantity is called the self distance D_{ii} of the subconductor

$D_{ii} = r_s e^{-0.25}$ if the subconductor is a solid cable, where r_s is the cable radius

$D_{ii} = GMR_s$ if the subconductor is a stranded cable, where GMR_s is the cable GMR from a table

The variable G in the transmission line equivalent circuits is called the **distributed shunt conductance**, and it models the resistance between the transmission line conductors through the insulation medium. Since the insulation medium used in the design of transmission lines is chosen particularly for its electrical insulation or non-conductive qualities, the resistance of the insulation medium is therefore very high (in the mega-Ohm range typically). If the resistance of the insulating medium is very high, then the conductance of the insulation medium will be very small. In fact the shunt conductance of most real world transmission lines is so small that it is assumed to be zero in transmission line calculations. In this course you should always assume the value of G to be zero, unless given a non-zero conductance value or specifically told otherwise.

The variable C in the transmission line equivalent circuits is called the **distributed shunt capacitance**, and it models the electric field interaction between the transmission line conductors through the insulation medium. Equations for both single phase and three phase transmission lines are given below.

Shunt Capacitance of a Single Phase Transmission Line

$$C = k_{C1} / \ln \left(\frac{GMD_{xy}}{\sqrt{mGMR_x \cdot mGMR_y}} \right) \quad k_{C1} = 4.47659 \times 10^{-8} F / mile \quad C_T = C \ell \quad F$$

$$= 2.781624 \times 10^{-11} F / meter$$

C = distributed shunt capacitance of the transmission line

C_T = total shunt capacitance of the transmission line

$$mGMR_x = \left[\prod_{i=1}^{n_x} \left(\prod_{j=1}^{n_x} D_{xixj}^* \right) \right]^{\frac{1}{n_x^2}} \quad mGMR_y = \left[\prod_{i=1}^{n_y} \left(\prod_{j=1}^{n_y} D_{yiyj}^* \right) \right]^{\frac{1}{n_y^2}}$$

$mGMR_x^*, mGMR_y^*$ = modified geometric mean radii of the two conductors

D_{xixj}^* = distance between subconductor i in conductor x and subconductor j in conductor x

D_{yiyj}^* = distance between subconductor i in conductor y and subconductor j in conductor y

if $i = j$ in D_{xixj}^* or D_{yiyj}^* , then the quantity is called the self distance D_{ii}^* of subconductor i

$D_{ii}^* = r_s$, where r_s is the cable radius (used for both solid and stranded cable subconductors)

Shunt Capacitance of a Three Phase Transmission Line

$$C = k_{C3} / \ln \left(\frac{GMD_{eq}}{mGMR} \right) \quad k_{C3} = 8.953181 \times 10^{-8} F / mile \quad C_T = C \ell \quad F$$

$$= 5.563249 \times 10^{-11} F / meter$$

C = distributed per phase shunt capacitance of the transmission line

C_T = total per phase shunt capacitance of the transmission line

$$mGMR = \left[\prod_{i=1}^n \left(\prod_{j=1}^n D_{ij}^* \right) \right]^{\frac{1}{n^2}}$$

$mGMR$ = modified geometric mean radius of one of the phase conductors

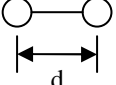
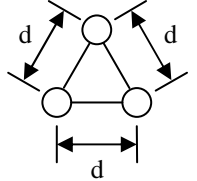
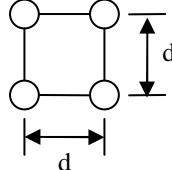
D_{ij}^* = distance between subconductor i in conductor x and subconductor j in conductor x

if $i = j$ in D_{ij}^* , then the quantity is called the self distance D_{ii}^* of subconductor i

$D_{ii}^* = r_s$, where r_s is the cable radius (used for both solid and stranded cable subconductors)

In the equations just tabulated for calculating the transmission line impedance parameters (R , L , C), the special variables n , GMR , and $mGMR$ were defined to account for conductor bundling in the calculations. Recall that conductor bundling is simply using more than one cable (or subconductor) in a transmission line conductor. Conductor bundling lowers net series resistance, increases current carrying capability, and reduces the electric field potential gradient at conductor surfaces. The equations defined for the GMR and $mGMR$ quantities are in the most general form which account for conductor bundling using n subconductors per conductor.

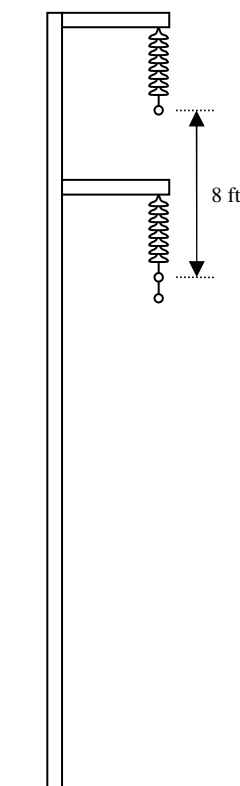
If and when conductor bundling is used in real world transmission lines, it is most often done using either two subconductors per conductor, three subconductors per conductor in an equilateral triangle configuration, or four subconductors per conductor in a square configuration. For these specific conductor bundling arrangements, shortcut equations for the GMR and $mGMR$ quantities have been derived from the general form equations and are listed below. These shortcut equations should be used instead of the general form equations.

Configuration	Bundling Diagram	GMR and $mGMR$ Equations
Single Cable (no bundling)		$GMR = GMR_s$ $mGMR = r_s$
Standard 2-Cable Bundle		$GMR = \sqrt{d \cdot GMR_s}$ $mGMR = \sqrt{d \cdot r_s}$
Standard 3-Cable Equilateral Bundle		$GMR = \sqrt[3]{d^2 \cdot GMR_s}$ $mGMR = \sqrt[3]{d^2 \cdot r_s}$
Standard 4-Cable Square Bundle		$GMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot GMR_s}$ $mGMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot r_s}$

For any other conductor bundling configuration, the general forms of the GMR and $mGMR$ equations that were previously presented must be used to calculate these quantities.

Example #1

A 1 ϕ transmission line uses a single ACSR *Piper* cable in one conductor and 2 ACSR *Waxwing* cables with 4" spacing in the other conductor as shown below. Cable data is also given below. Calculate the distributed series resistance, distributed series inductance, and distributed shunt capacitance of this transmission line.



Code Name	Strands (Al/St)	Al Area (cmil)	diameter (in)	radius (ft)	GMR (ft)	R (Ω /mile) 60Hz, 20°C	$\Delta R/\Delta T$	Rated Amps
Waxwing	18/1	266800	0.609	0.0254	0.0198	0.3488	0.0014	465
Piper	30/7	300000	0.700	0.0292	0.0241	0.3077	0.0012	485

Distributed series resistance of the transmission line:

$$R = \frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y} = \frac{0.3077}{1} + \frac{0.3488}{2} = 0.4821 \Omega / \text{mi}$$

Distributed series inductance of the transmission line:

$$GMR_x = GMR_{sx} = 0.0241 \text{ ft} \quad (\text{no bundling})$$

$$GMR_y = \sqrt{d \cdot GMR_{sy}} = \sqrt{\left(\frac{4}{12}\right)(0.0198)} = 0.08124 \text{ ft} \quad (\text{standard 2-cable bundle})$$

$$D_{x1y1} = 8 \text{ ft}, \quad D_{x1y2} = 8 + \frac{4}{12} = 8.33 \text{ ft}, \quad GMD_{xy} = \sqrt{D_{x1y1} D_{x1y2}} = 8.16497 \text{ ft}$$

$$L = k_{L1} \cdot \ln \left(GMD_{xy} / \sqrt{GMR_x \cdot GMR_y} \right) = 3.358891 \text{ mH} / \text{mile}$$

Distributed shunt capacitance of the transmission line:

$$mGMR_x = r_{sx} = 0.0292 \text{ ft} \quad (\text{no bundling})$$

$$mGMR_y = \sqrt{d \cdot r_{sy}} = \sqrt{\left(\frac{4}{12}\right)(0.0254)} = 0.092014 \text{ ft} \quad (\text{standard 2-cable bundle})$$

$$C = k_{C1} / \ln \left(GMD_{xy} / \sqrt{mGMR_x \cdot mGMR_y} \right) = 8.847802 \text{ nF} / \text{mi}$$

Example #2

A 1 ϕ power cable is constructed using two solid copper wires enclosed and separated by a special type of polyvinyl insulating material. Each conductor of the power cable has a radius of 0.135" and a distributed series resistance of 0.0172 Ω per 1000ft. The two conductors are separated by a center to center distance of 0.42". The polyvinyl insulating material used has an inductance constant that is 1.08 times greater than that of air, and has a capacitance constant that is 11.25 times greater than that of air. Calculate the distributed series resistance, distributed series inductance, and distributed shunt capacitance of this transmission line. Express these results in units per 1000ft.

Distributed series resistance of the transmission line:

$$R = \frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y} = \frac{0.0172}{1} + \frac{0.0172}{1} = 0.0344 \Omega / \text{kft}$$

Distributed series inductance of the transmission line:

$$GMR_x = GMR_y = e^{-0.25} r_s = e^{-0.25} (0.135 \text{ in}) = 0.10514 \text{ in} \quad (\text{no bundling})$$

$$GMD_{xy} = D_{x1y1} = 0.42 \text{ in}$$

$$L = 1.08 \cdot k_{L1} \cdot \ln \left(GMD_{xy} / \sqrt{GMR_x \cdot GMR_y} \right) = (962.8888 \mu\text{H} / \text{mi}) \left(\frac{1 \text{ mi}}{5.280 \text{ kft}} \right) = 182.3653 \mu\text{H} / \text{kft}$$

Distributed shunt capacitance of the transmission line:

$$mGMR_x = mGMR_y = r_s = 0.135 \text{ in} \quad (\text{no bundling})$$

$$C = 11.25 \cdot k_{C1} / \ln \left(GMD_{xy} / \sqrt{mGMR_x \cdot mGMR_y} \right) = (443.7227 \text{ nF} / \text{mi}) \left(\frac{1 \text{ mi}}{5.280 \text{ kft}} \right) = 84.0384 \text{ nF} / \text{kft}$$

Example #3

A 3 ϕ power cable is constructed using three solid copper wires enclosed and separated by a special type of polyvinyl insulating material. Each conductor of the power cable has a radius of 0.135" and a distributed series resistance of 0.0172 Ω per 1000ft. The three conductors are arranged in a side-by-side flat configuration with a center to center distance of 0.42" between adjacent conductors. The polyvinyl insulating material used has an inductance constant that is 1.08 times greater than that of air, and has a capacitance constant that is 11.25 times

greater than that of air. Calculate the distributed per phase series resistance, distributed per phase series inductance, and distributed per phase shunt capacitance of this transmission line. Express these results in units per 1000ft.

Distributed per phase series resistance of the transmission line:

$$R = \frac{R_s}{n} = \frac{0.0172}{1} = 0.0172 \, \Omega / \text{kft}$$

Distributed per phase series inductance of the transmission line:

$$GMR = e^{-0.25} r_s = e^{-0.25} (0.135 \text{ in}) = 0.10514 \text{ in} \quad (\text{no bundling})$$

$$GMD_{eq} = \sqrt[3]{D_{AB} D_{BC} D_{CA}} = \sqrt[3]{(0.42)(0.42)(0.84)} = 0.52917 \text{ in}$$

$$L = 1.08 \cdot k_{L3} \cdot \ln(GMD_{eq} / GMR) = (561.7613 \, \mu\text{H} / \text{mi}) \left(\frac{1 \text{ mi}}{5.280 \text{ kft}} \right) = 106.3942 \, \mu\text{H} / \text{kft}$$

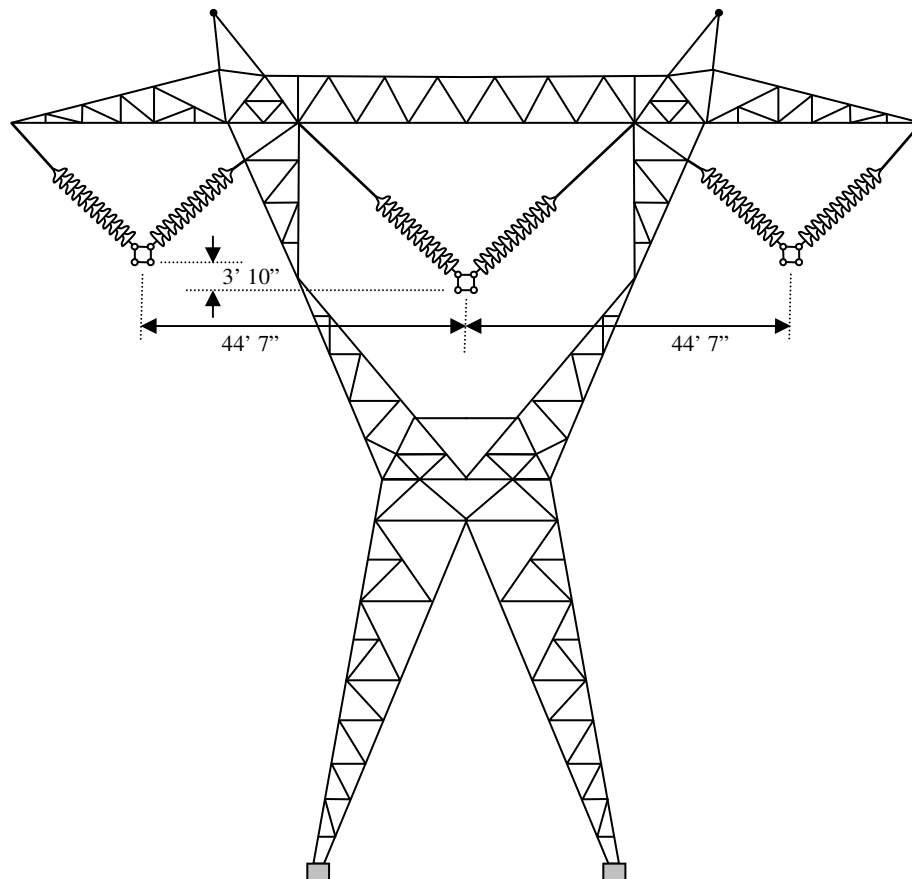
Distributed per phase shunt capacitance of the transmission line:

$$mGMR = r_s = 0.135 \text{ in} \quad (\text{no bundling})$$

$$C = 11.25 \cdot k_{C3} / \ln(GMD_{eq} / mGMR) = (737.3437 \text{ nF} / \text{mi}) \left(\frac{1 \text{ mi}}{5.280 \text{ kft}} \right) = 139.6484 \text{ nF} / \text{kft}$$

Example #4

A 3 ϕ 100 mile transmission line designed to operate at 765KV uses four ACSR *Bluebird* cables per phase in 26 inch square bundles as shown in the support tower diagram below. Cable data is also given below. Calculate the Pi Model L-N equivalent circuit parameters for this transmission line.



Code Name	Strands (Al/St)	Al Area (cmil)	diameter (in)	radius (ft)	GMR (ft)	R (Ω /mile) 60Hz, 20°C	$\Delta R/\Delta T$	Rated Amps
Bluebird	84/19	2156000	1.762	0.0734	0.0586	0.0476	0.00019	1665

Distributed per phase series resistance of the transmission line:

$$R = \frac{R_s}{n} = \frac{0.0476}{4} = 0.0119 \, \Omega / mi$$

Distributed per phase series inductance of the transmission line:

$$GMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot GMR_s} = \sqrt[4]{\sqrt{2} \left(\frac{26}{12}\right)^3 (0.0586)} = 0.958180 \, ft \quad (4\text{-cable square bundle})$$

$$D_{AB} = D_{BC} = \sqrt{\left(44 + \frac{7}{12}\right)^2 + \left(3 + \frac{10}{12}\right)^2} = 44.74783 \, ft, \quad D_{CA} = 2\left(44 + \frac{7}{12}\right) = 89.166 \, ft$$

$$GMD_{eq} = \sqrt[3]{D_{AB} D_{BC} D_{CA}} = 56.30956 \, ft$$

$$L = k_{L3} \cdot \ln(GMD_{eq} / GMR) = 1.3112 \, mH / mi$$

Distributed per phase shunt capacitance of the transmission line:

$$mGMR = \sqrt[4]{\sqrt{2} \cdot d^3 \cdot r_s} = \sqrt[4]{\sqrt{2} \left(\frac{26}{12}\right)^3 (0.0734)} = 1.01367 \, ft \quad (4\text{-cable square bundle})$$

$$C = k_{C3} / \ln(GMD_{eq} / mGMR) = 22.2866 \, nF / mi$$

Pi Model L-N Equivalent Circuit Parameters for the Transmission Line:

$$\vec{Z}_{TL} = \vec{Z} \ell = (R + j120\pi L) \ell$$

$$= \left[(0.0119 \, \Omega / mi) + j120\pi (1.3112 \times 10^{-3} \, H / mi) \right] (100 \, mi) = 49.44387 \angle 88.621^\circ \, \Omega$$

$$\vec{Y}_{TL} = \frac{1}{2} \vec{y} \ell = \frac{1}{2} (G + j120\pi C) \ell$$

$$= \frac{1}{2} \left[0 + j120\pi (22.2866 \times 10^{-9} \, F / mi) \right] (100 \, mi) = 4.20093 \times 10^{-4} \angle 90^\circ \, \mathfrak{S}$$

It was previously stated that transmission lines are modeled using either the **Series Impedance Model** equivalent circuit or the **Pi Model** equivalent circuit. There is one more type of transmission line model, called the **Long Length Model**, that can also be used.

The **Long Length Model** is very unique and different from the other two models. It is derived by writing and solving differential equations that model the true distributed nature of the four transmission line phenomena as a function of the distance along the length of the transmission line. Consequently the Long Length Model allows calculation of voltage and current at any point along the transmission line length between its end points, and it is therefore called a **distributed parameter model**. The Series Impedance Model and the Pi Model are only capable of calculating transmission line end point voltage and current, and are therefore called **lumped parameter models**. The Long Length Model has the following equivalent circuit and equations.

Transmission Line Long Length Model Equivalent Circuit and Equations	
	$\begin{cases} \vec{Z} = R + j120\pi L \\ \vec{y} = G + j120\pi C \end{cases} \quad \begin{cases} \vec{\gamma} = \sqrt{\vec{Z} \cdot \vec{y}} \\ \vec{Z}_c = \sqrt{\vec{Z} / \vec{y}} \end{cases}$ $\vec{Z}_{TL} = \vec{Z}_c \sinh(\vec{\gamma} \ell) \quad \vec{Y}_{TL} = \frac{\cosh(\vec{\gamma} \ell) - 1}{\vec{Z}_c \sinh(\vec{\gamma} \ell)}$ $\begin{cases} \vec{V}(x) = [\cosh(\vec{\gamma} x)] \vec{V}_R + [\vec{Z}_c \sinh(\vec{\gamma} x)] \vec{I}_R \\ \vec{I}(x) = [\vec{Z}_c^{-1} \sinh(\vec{\gamma} x)] \vec{V}_R + [\cosh(\vec{\gamma} x)] \vec{I}_R \end{cases}$ $x = 0 \text{ at the receiving end, } x = \ell \text{ at the sending end}$

Notice that the Long Length Model equivalent circuit looks exactly the same as the Pi Model equivalent circuit. However, notice that its parameters are defined very differently, using hyperbolic functions. Also notice that in the equations for the Long Length Model, two new intermediate quantities are defined. These quantities are called the propagation constant ($\bar{\gamma}$) and the characteristic impedance ($\bar{Z}_c$) of the transmission line, and they are calculated from the distributed series impedance and distributed shunt admittance.

Example #5

For the transmission line in Example #4, calculate the Long Length Model L-N equivalent circuit parameters.

$$\ell = 100 \text{ mi}, \quad \bar{\gamma} = \sqrt{\bar{z} \cdot \bar{y}} = 2.038187 \times 10^{-3} \angle 89.310^\circ / \text{mi}, \quad \bar{Z}_c = \sqrt{\bar{z} / \bar{y}} = 242.587481 \angle -0.690^\circ \Omega$$

$$\bar{Z}_{TL} = \bar{Z}_c \sinh(\bar{\gamma}\ell) = 49.102347 \angle 88.630^\circ \Omega$$

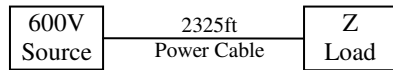
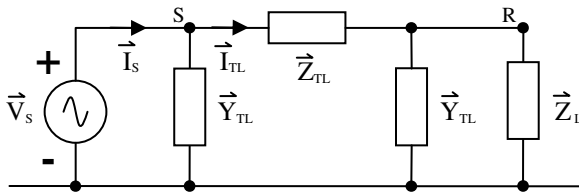
$$\bar{Y}_{TL} = \frac{\cosh(\bar{\gamma}\ell) - 1}{\bar{Z}_c \sinh(\bar{\gamma}\ell)} = 4.215532 \times 10^{-4} \angle 89.995^\circ \text{ S}$$

Example #6

A 600V source is connected through 2325ft of the power cable defined in Example #2 to a load impedance of value $50 \angle 65^\circ \Omega$. Calculate the load voltage using the Long Length Model for the power cable.

Step 1: Visualize

A block diagram is drawn below.


Step 2: Equivalent Circuit

Step 3: Circuit Parameters

$$\bar{V}_S = 600 \angle 0^\circ \text{ V}, \quad \bar{Z}_L = 50 \angle 65^\circ \Omega$$

$$\ell = 2325 \text{ ft} = 2.325 \text{ kft}, \quad R = 0.0344 \Omega / \text{kft}, \quad L = 182.3653 \mu\text{H} / \text{kft}, \quad C = 84.0384 \text{ nF} / \text{kft}$$

$$\bar{z} = R + j120\pi L = 0.076876 \angle 63.418^\circ \Omega / \text{kft}, \quad \bar{y} = G + j120\pi C = 1.008461 \times 10^{-5} \angle 90^\circ \text{ S} / \text{kft}$$

$$\bar{\gamma} = \sqrt{\bar{z} \cdot \bar{y}} = 8.804916 \times 10^{-4} \angle 76.709^\circ / \text{kft}, \quad \bar{Z}_c = \sqrt{\bar{z} / \bar{y}} = 87.310442 \angle -13.291^\circ \Omega$$

$$\bar{Z}_{TL} = \bar{Z}_c \sinh(\bar{\gamma}\ell) = 0.178737 \angle 63.418^\circ \Omega, \quad \bar{Y}_{TL} = \frac{\cosh(\bar{\gamma}\ell) - 1}{\bar{Z}_c \sinh(\bar{\gamma}\ell)} = 1.172336 \times 10^{-5} \angle 90^\circ \text{ S}$$

Step 4: Circuit Analysis

$$\bar{Z}_{eqR} = (\bar{Y}_{TL} + \bar{Z}_L^{-1})^{-1} = 50.026575 \angle 64.986^\circ \Omega, \quad \bar{Z}_{eqS} = \left[\bar{Y}_{TL} + (\bar{Z}_{TL} + \bar{Z}_{eqR})^{-1} \right]^{-1} = 50.232035 \angle 64.966^\circ \Omega$$

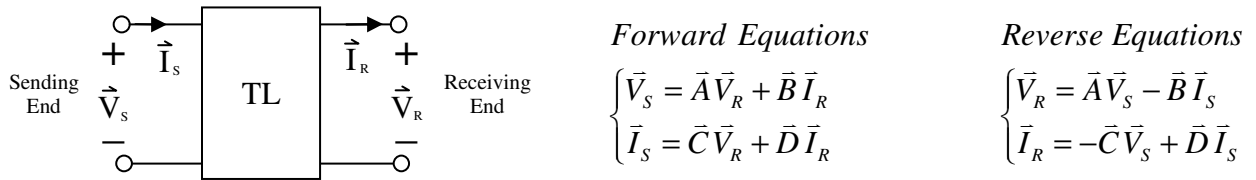
$$\bar{I}_S = \bar{V}_S \bar{Z}_{eqS}^{-1} = 11.944569 \angle -64.966^\circ \text{ A}, \quad \bar{I}_{TL} = \bar{I}_S - \bar{Y}_{TL} \bar{V}_S = 11.950943 \angle -64.980^\circ \text{ A}$$

$$\bar{V}_R = \bar{Z}_{eqR} \bar{I}_{TL} = 597.864723 \angle -0.006^\circ \text{ V}$$

Step 5: Final Results

$$\text{Load Voltage} = |\bar{V}_R| = 597.865 \text{ V}$$

Another method of doing transmission line end point voltage and current calculations is to use a 2-port network model and 2-port network equations for the transmission line. With this method, the transmission line is modeled by a black-box instead of an equivalent circuit, and the end point voltage/current calculations are done using algebraic equations instead of using circuit analysis. The general 2-port network model for a transmission line and its algebraic equations are given below.



The coefficients of the 2-port network model algebraic equations are called the ABCD parameters of the transmission line. The ABCD parameters numerically represent the physical characteristics and electrical phenomena of the transmission line. There are three specific ways to define the ABCD parameters of a transmission line that are based on the voltage/current behavior of the same three different equivalent circuit models previously defined. These three definitions of the ABCD parameters are listed in the table below.

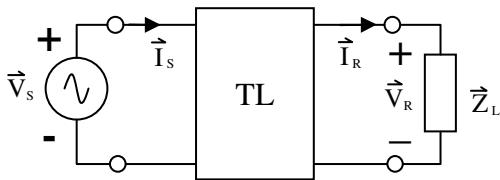
Short Length Model (same as the Series Impedance Model)	$\bar{A} = \bar{D} = 1, \quad \bar{B} = \bar{Z}_{TL}, \quad \bar{C} = 0$
Medium Length Model (same as the Pi Model)	$\bar{A} = \bar{D} = 1 + \bar{Z}_{TL} \bar{Y}_{TL}, \quad \bar{B} = \bar{Z}_{TL}, \quad \bar{C} = \bar{Y}_{TL} (2 + \bar{Z}_{TL} \bar{Y}_{TL})$
Long Length Model	$\bar{A} = \bar{D} = \cosh(\bar{\gamma} \ell), \quad \bar{B} = \bar{Z}_c \sinh(\bar{\gamma} \ell), \quad \bar{C} = \bar{Z}_c^{-1} \sinh(\bar{\gamma} \ell)$

Using the 2-port network model for a transmission line is very useful and convenient when calculations are being done for only a transmission line in a power system instead of doing an analysis of the entire power system. The 2-port network model allows transmission line end point voltage/current calculations to be done without drawing an equivalent circuit for the system and without doing circuit analysis.

Example #7

Rework Example #6 using the 2-port network model for the transmission line instead of the equivalent circuit model.

Step 2: Equivalent Circuit



Step 3: Circuit Parameters

$$\bar{A} = \bar{D} = \cosh(\bar{\gamma} \ell) = 0.999998 \angle 0.0001^\circ, \quad \bar{B} = \bar{Z}_c \sinh(\bar{\gamma} \ell) = 0.178737 \angle 63.418^\circ$$

$$\bar{C} = \bar{Z}_c^{-1} \sinh(\bar{\gamma} \ell) = 2.344670 \times 10^{-8} \angle 90^\circ$$

Step 4: Circuit Analysis

$$\vec{V}_S = \bar{A} \vec{V}_R + \bar{B} \vec{I}_R = \bar{A} \vec{V}_R + \bar{B} (\vec{V}_R \bar{Z}_L^{-1}) = (\bar{A} + \bar{B} \bar{Z}_L^{-1}) \vec{V}_R \rightarrow \vec{V}_R = \vec{V}_S (\bar{A} + \bar{B} \bar{Z}_L^{-1})^{-1} = 597.864723 \angle -0.006^\circ V$$

Step 5: Final Results

$$\text{Load Voltage} = |\vec{V}_R| = 597.865 V$$